

		2 2
3	C	(As the rifle was aimed at pigeon which was 60 m high) $\theta = \tan^{-1} \frac{60}{170}$ Consider horizontal motion: (to hit the pigeon, the bullet would have travelled 170 m horizontally) $t = \frac{s_x}{u_x} = \frac{170}{300 \cos \theta} = 0.60 \text{ s}$
4	C	Change in momentum = mass × change in velocity